LMS Data Flow Diagrams

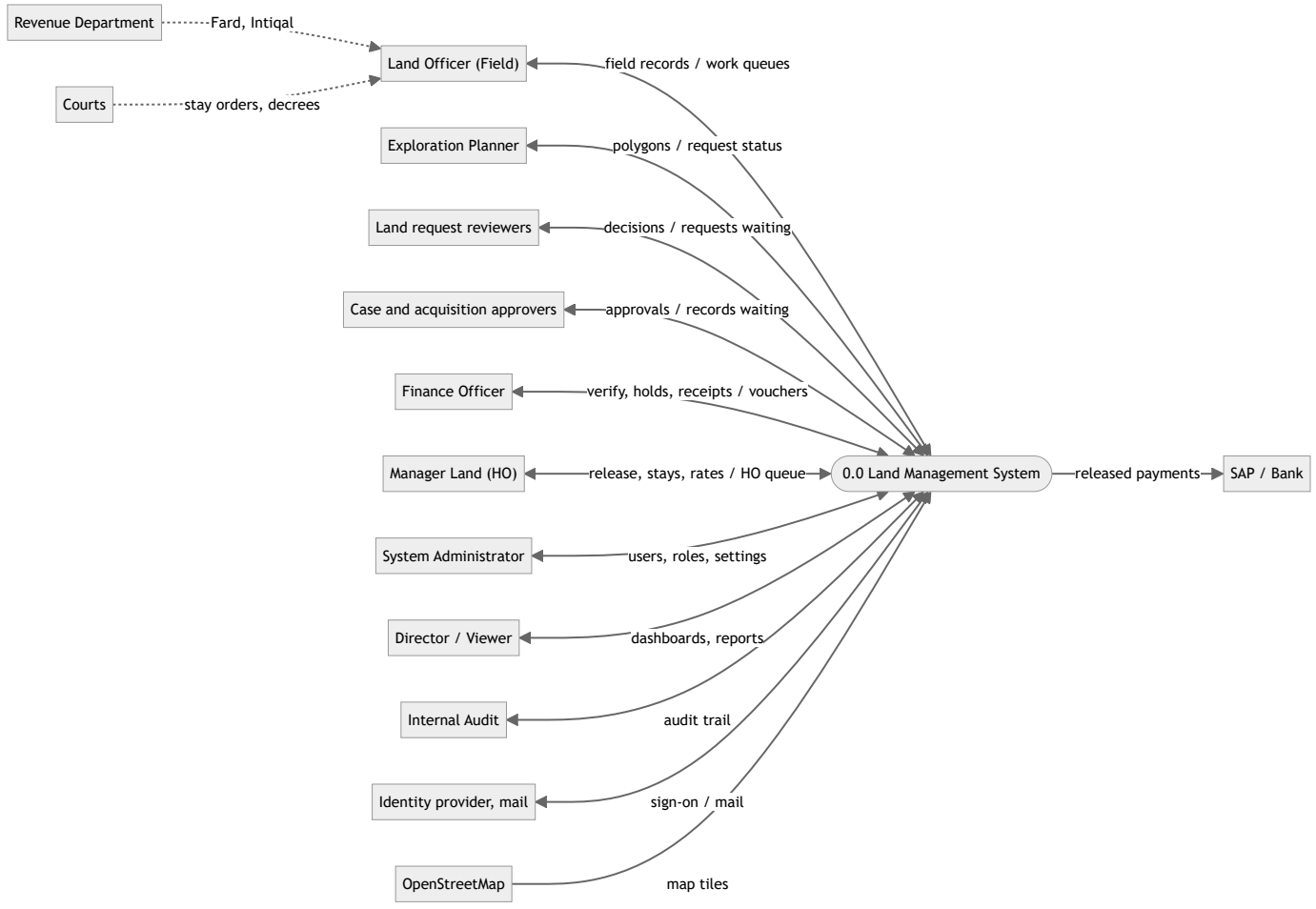
Mari Energies Land Management System. The diagrams show how data moves between people, outside parties, the system's processes and its data stores. They describe the system as built on 3 Nov 2026 (database migration 0062).

How to read them

Shape	Meaning
Rectangle	External entity: a person or outside party
Rounded box with a number	Process: work the system does (1.0, 2.0 ...)
Cylinder with D-number	Data store: where the data is kept (D1, D2 ...)
Arrow label	The data that moves
Dotted arrow	Happens outside the system, or by hand

1. Level 0: Context diagram

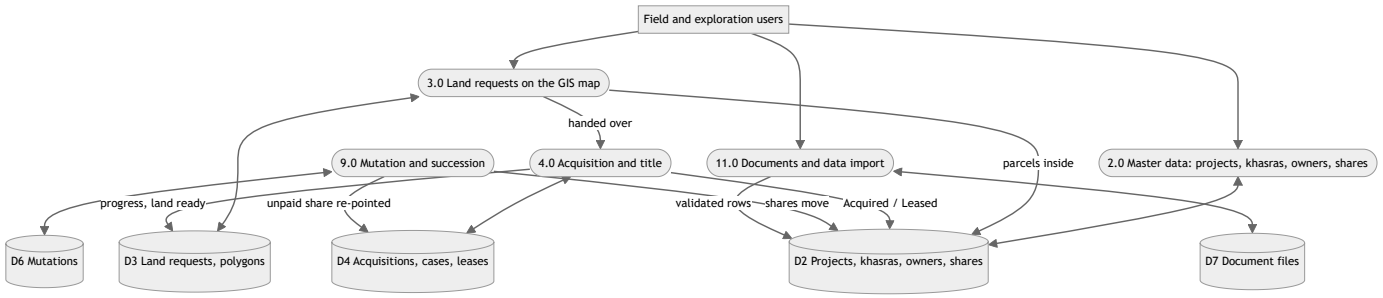
The whole system as one process, with everyone who gives it data or takes data from it. The full list of what each one gives and gets is in section 5.



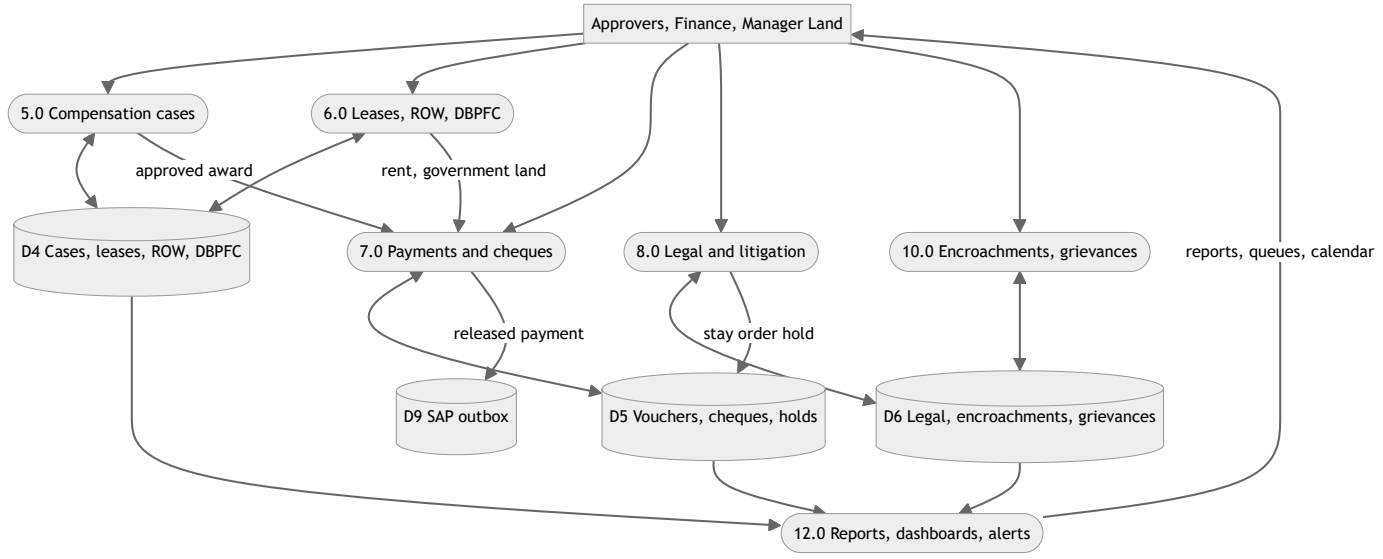
2. Level 1: Main processes and data stores

Shown in two parts. **Every process** also writes each change to **D8 Audit trail**, and every request first passes **1.0 Sign in and access control** (see 3.1).

2a. Land side



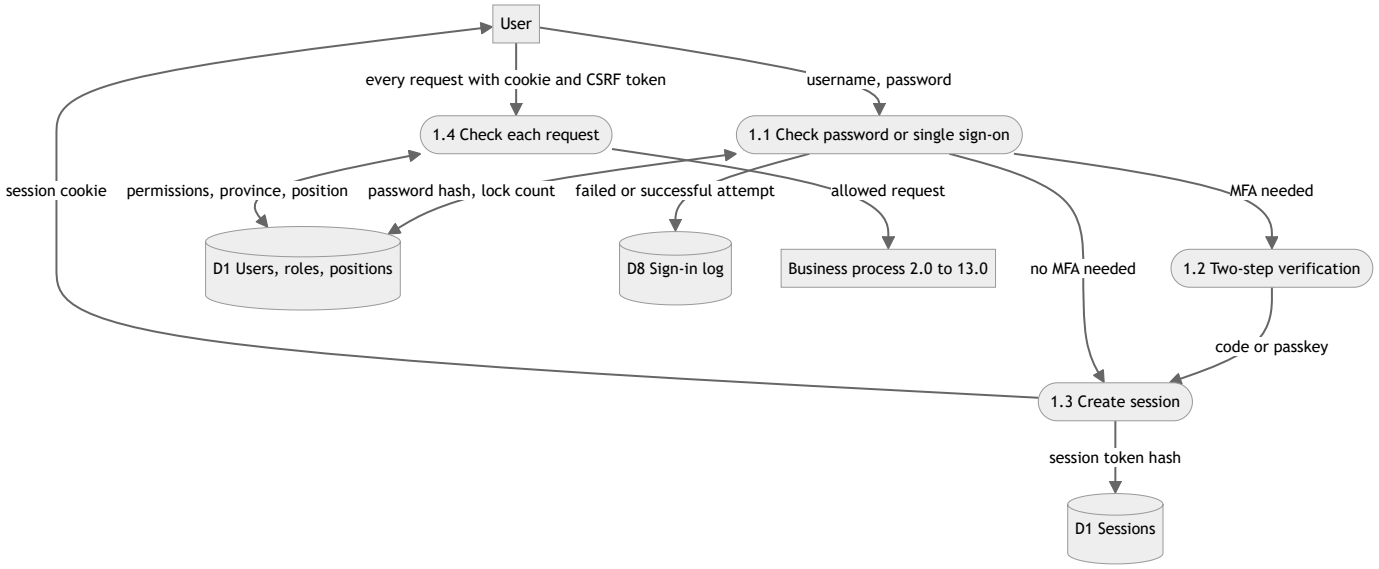
2b. Money and control side



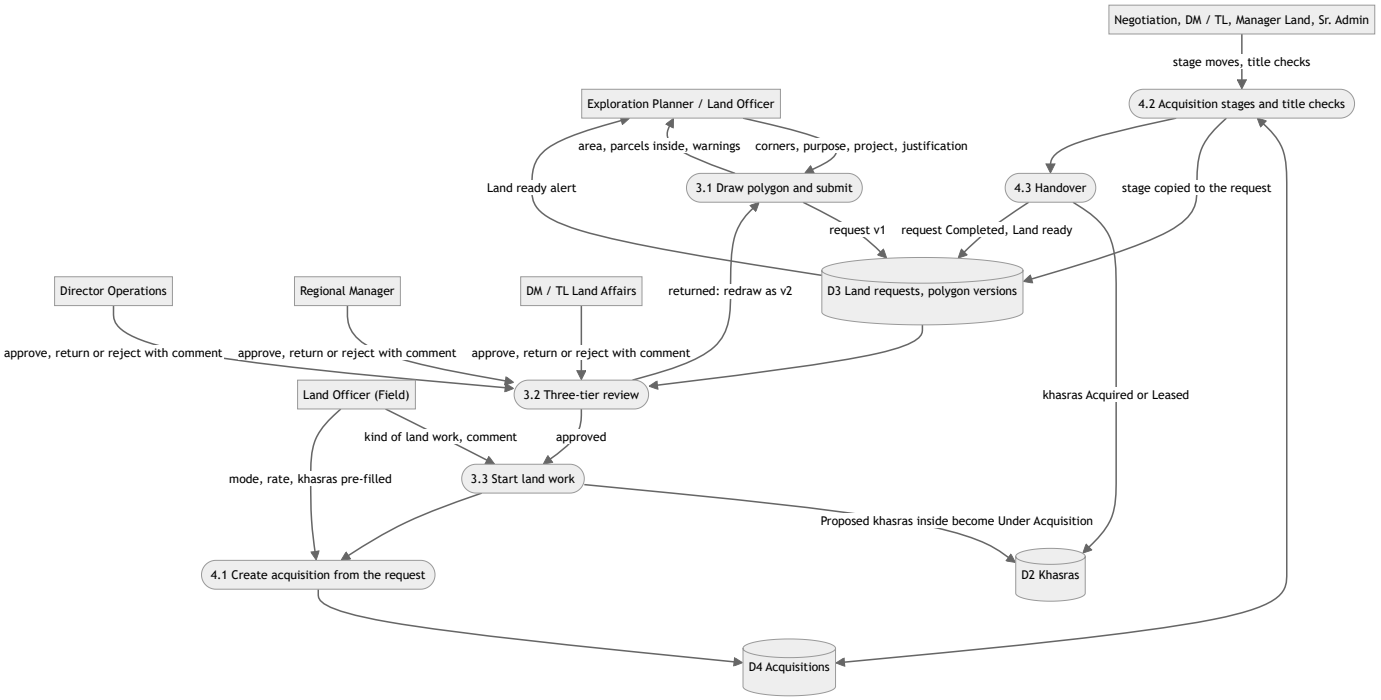
3. Level 2 flows

3.1 Process 1.0: Sign in and access control

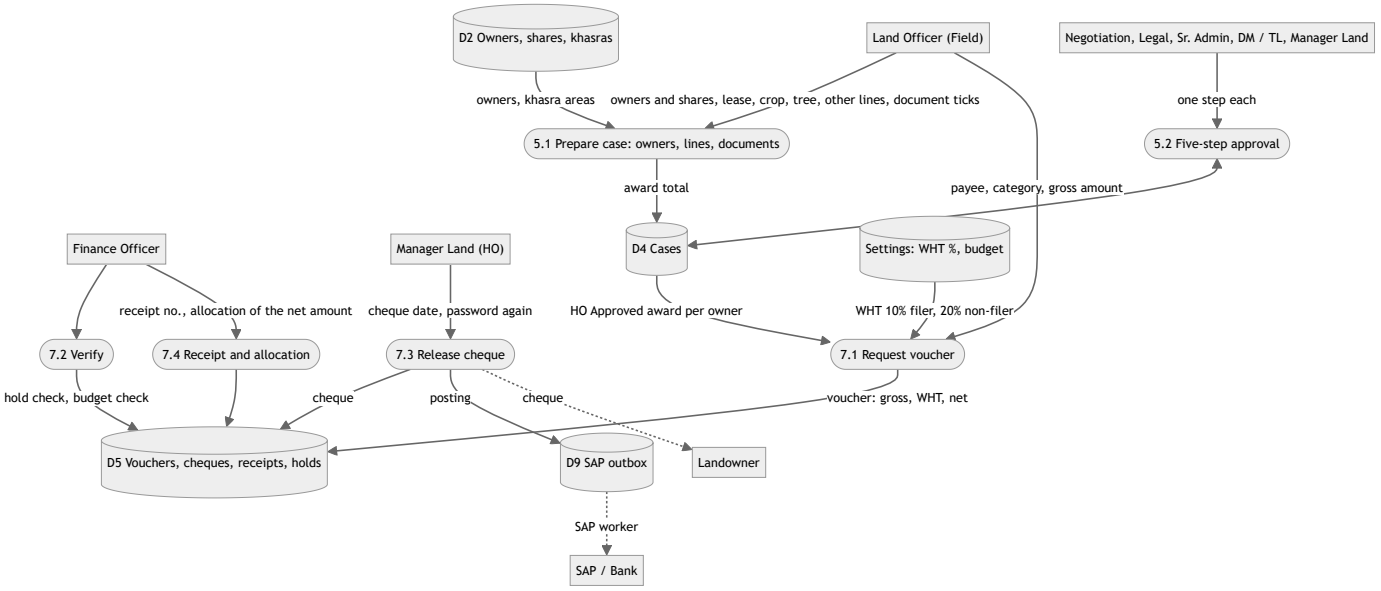
Every request runs as the signed-in user inside one database transaction. The database itself checks the user's province, permission and approval position before any data is read or changed.



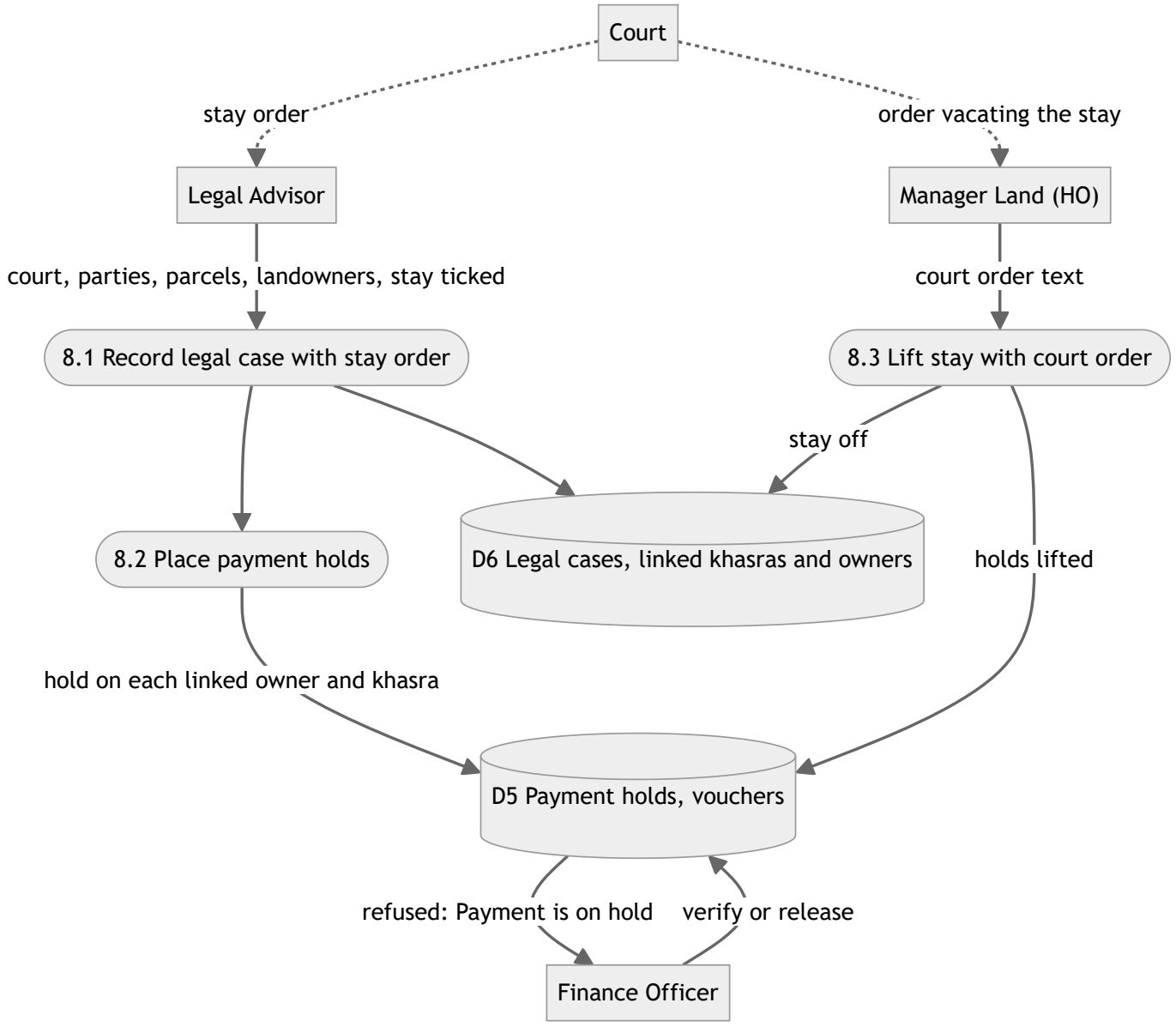
3.2 Processes 3.0 and 4.0: Land request to land ready



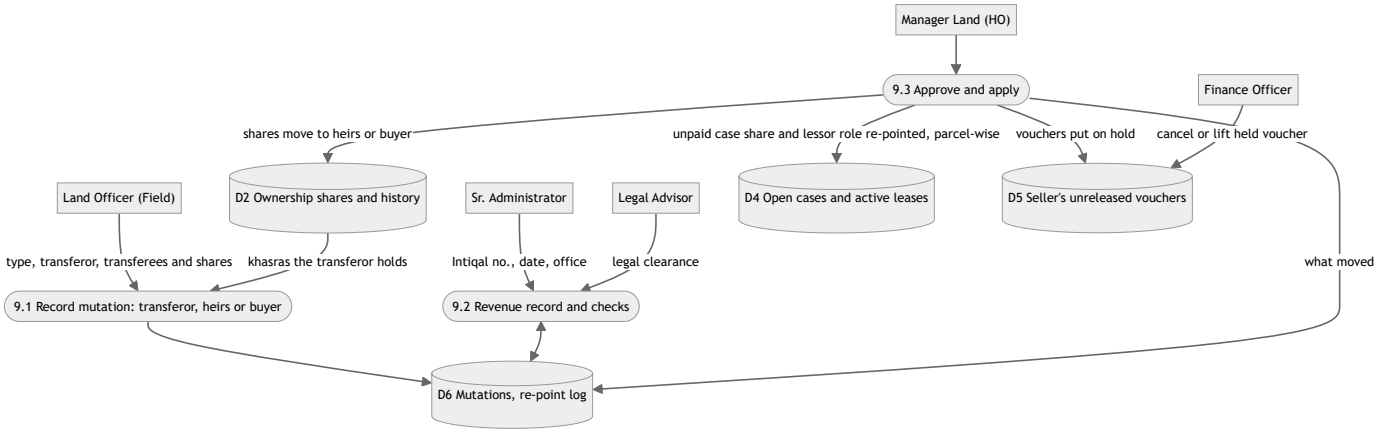
3.3 Processes 5.0 and 7.0: Compensation to cheque



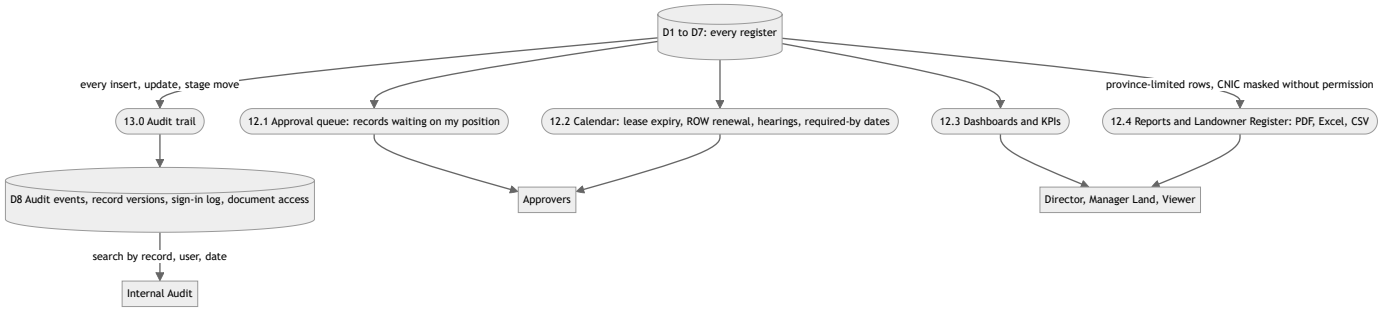
3.4 Process 8.0: Court stay stops payment



3.5 Process 9.0: Mutation moves shares and payments



3.6 Processes 12.0 and 13.0: Reports, alerts and audit



4. Data stores

Store	Holds	Main tables
D1	Users, roles, permissions, approval positions, provinces, sessions, MFA	`lms.app_user`, `lms.app_user_group`, `lms.app_user_province`, `iam.role`, `iam.role_permission`, `iam.user_role`, `lms.login_session`, `iam.mfa_*`
D2	Projects, administrative hierarchy, khasras, landowners, ownership shares	`lms.project`, `lms.admin_unit`, `lms.survey`, `lms.landowner`, `lms.survey_ownership_share`, `lms.survey_ownership_history`
D3	Land requests, versioned polygons	`lms.land_request`, `lms.land_request_area`
D4	Acquisitions, title checks, compensation cases and lines, leases, ROW corridors, DBPFC estimates, budgets	`lms.acquisition`, `lms.acquisition_title_check`, `lms.compensation_case`, `lms.case_*`, `lms.lease`, `lms.row_corridor`, `lms.dbpfc_estimate`, `lms.budget`
D5	Vouchers, cheques, reversals, receipts, allocations, payment holds, reconciliation	`lms.payment_request`, `lms.payment_release`, `lms.payment_release_reversal`, `lms.payment_receipt`, `lms.payment_allocation`, `lms.payment_hold`, `lms.reconciliation_run`
D6	Legal cases and hearings, mutations and re-point log, encroachments, grievances	`lms.legal_case`, `lms.legal_hearing`, `lms.mutation`, `lms.mutation_repoint`, `lms.encroachment`, `lms.grievance`
D7	Uploaded files (stored by content hash), KML layers, import batches	`lms.document`, `lms.project_kml_import`, `lms.import_batch` + the document folder on disk
D8	Audit events, record versions, sign-in events, document access log	`audit.audit_event`, `audit.record_version`, `audit.auth_event`, `lms.document_access_log`
D9	Released payments waiting to be posted to SAP	`lms.integration_outbox`
—	Workflow stages, who approves each step, settings	`lms.workflow_stage`, `lms.workflow_step_group`, `lms.system_setting`

5. External entities

Entity	Gives the system	Gets from the system
Land Officer (Field)	Projects, khasras, owners, shares, cases, vouchers, mutations, encroachments, grievances, documents	Work queues, alerts, calendar
Exploration Planner	Land request polygons	Request status, Land ready
Land request reviewers (Director Operations, Regional Manager, DM / TL Land Affairs)	Approve, return, reject	Requests waiting on them
Case and acquisition approvers (Negotiation, Legal, Sr. Admin, DM / TL Field, Manager Land)	One approval step each	Records waiting on their position
Finance Officer	Verification, holds, receipts, cancellations	Vouchers, cheques, reconciliation
Manager Land (HO)	Cheque release, lifting stays, rate fixing, budgets	Head-office queue, dashboards
Director / Viewer	—	Dashboards, reports
Internal Audit	—	Audit trail, sign-in log
System Administrator	Users, roles, positions, settings, land units	User and session lists
Revenue Department (outside the system)	Fard, Intiqal, revenue records, entered by the Land Officer	—
Courts (outside the system)	Stay orders and decrees, entered by Legal / Manager Land	—
SAP / Bank	—	Postings of released payments (D9)
Identity provider, mail server	Single sign-on identity	Invitation and password-reset mail
OpenStreetMap	Map tiles	—